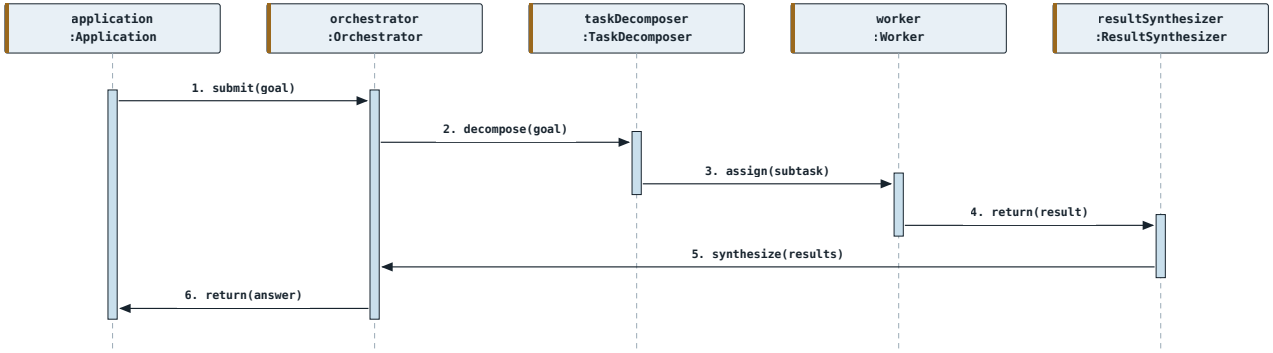
W04 Orchestrator-workers

Workflow / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Generate and delegate subtasks dynamically.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE	
CONTEXT	A lead agent can generate tasks rather than call a fixed pipeline.
TRIGGER	The orchestrator decomposes a goal or dispatches a subtask.
STRUCTURE	Task plan, worker selection, delegation envelope, and synthesis step.
EFFECT	Synthesized output accounts for generated task results and delegated authority.

CONTRACT / REQUIRED BEHAVIOR	
REQUIRES	Workers advertise capabilities and share a task/result protocol.
ENSURES	Every dispatched task has a result or failure and the final answer links them.
INVARIANT	Dynamic dispatch cannot create unbounded uncounted worker expansion.
VIOLATION	Unsplittable, failed, or duplicate work is reflected in synthesis.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.